

SPAT-VR: A Holistic and Extensible Framework for VR Project Management

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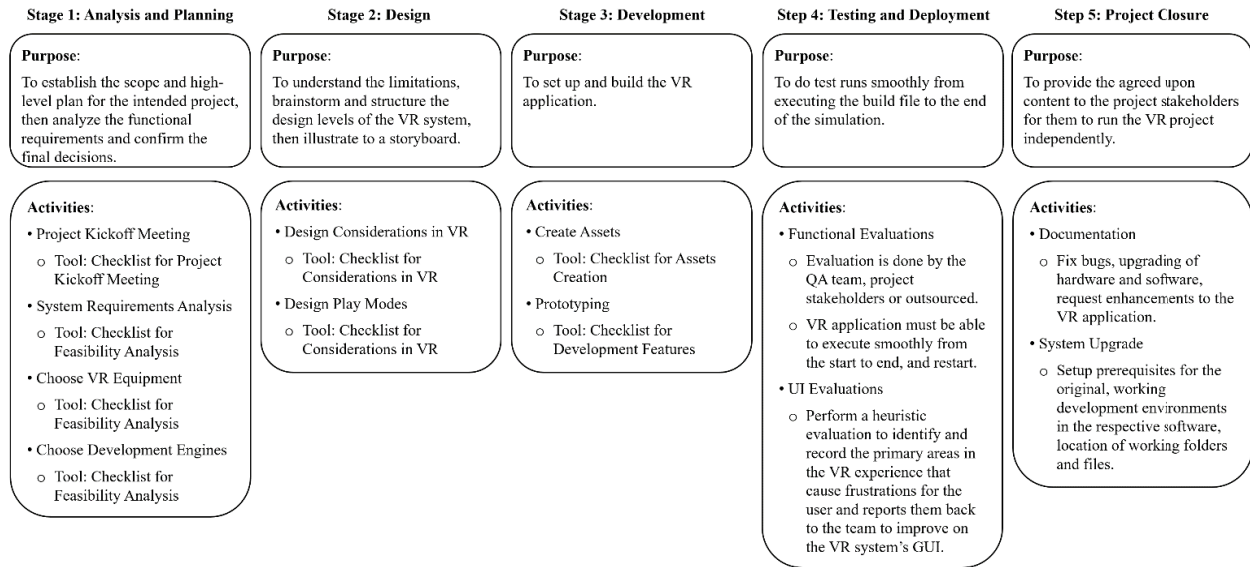


Figure 1: An overview of the proposed SPAT framework for VR project management.

ABSTRACT

With the increasing availability of VR content and affordability of consumer VR hardware systems, VR has been applied in various domains. However, to our knowledge, at present there are no existing frameworks applicable project management of VR applications. In this paper, we present SPAT-VR: a holistic and extensible framework for VR project management. The proposed framework has been successfully deployed to complete a number of VR projects. A user survey has been conducted with all positive results received on the proposed framework.

Keywords: Virtual reality, project management.

1 INTRODUCTION

Virtual Reality (VR) provides an immersive experience to users by generating and displaying virtual contents to the user. With the increasing availability of VR content and affordability of consumer VR hardware systems, VR has been applied in various domains [1], [2], [3].

Project management is defined as “the use of specific knowledge, skills, tools and techniques to deliver something of value to people.”¹. Regardless of industries, implementation of proven

project management practices has resulted in a 2.5 times success rate of projects. However, to our knowledge, at present there are no existing frameworks that are applicable to the project management of the entire life cycle of building VR applications. Although there are some systematic literature and scoping reviews, most of them are focused on only the design phase of VR projects that are targeted towards specific applications.

In this paper, we present a holistic and extensible framework for VR project management. As our framework comprises 4 key elements, namely Stages, Purposes, Activities and Tools, it is named as SPAT-VR. The SPAT-VR framework is based on years of VR project management experiences in different domains, therefore, it will provide holistic guidance for VR project managers. Additionally, it can be easily extended by incorporating newly developed technologies to cater for the dynamic advancements of VR. SPAT-VR has been successfully utilized to guide the development of VR projects.

2 SPAT-VR: OUR PROPOSED HOLISTIC AND EXTENSIBLE FRAMEWORK FOR VR PROJECT MANAGEMENT

Our SPAT-VR framework comprises 4 key elements: Stages, Purposes, Activities and Tools. For each VR project, its life cycle is divided into five stages, namely Analysis and Planning, Design, Development, Testing and Deployment, and Project Closure (Figure 1). Each stage serves different purposes and includes a set of activities and tools to be conducted.

¹ <https://www.pmi.org/about/learn-about-pmi/what-is-project-management>

2.1 Stage 1: Analysis and Planning

2.1.1 Purpose

The Analysis and Planning stage establishes the scope and high-level plan for the project. Resources, budget, time, milestones, deliverables, end users are considered here for feasibility assessment. Eventually all the discussed factors that are agreed upon by all stakeholders will be documented in the project plan.

2.1.2 Activities and Tools

There are a few activities engaged in this stage, including: Project Kickoff Meeting, System Requirements Analysis, Choose VR Equipment, and Choose Development Engines. For each activity, tools such as 5W1H system (Table 1 in supplementary material), checklist for feasibility analysis (Table 2 in supplementary material), can be used to facilitate the activity.

2.2 Stage 2: Design

2.2.1 Purpose

The Design Stage is to convert the conducted requirements analysis into level designs and storyboard of the VR system. It involves the design thinking and solutions (i.e., operations, training procedures, maintenance plans) that will satisfy the functional requirements of the proposed VR system.

2.2.2 Activities and Tools

Two key activities are conducted at this stage, including Design Considerations in VR, and Design Play Modes. The tools for these two activities are summarized in Table 3 of the supplementary material.

2.3 Stage 3: Development

2.3.1 Purpose

This stage involves the development of digital assets and the developer begins programming VR application for various design tasks according to the design documents and outlined specifications.

2.3.2 Activities and Tools

Two key activities are conducted at this stage, Create Assets, and Prototype and Development. The tools for these two activities are summarized in Table 4 of the supplementary material.

2.4 Stage 4: Testing and Deployment

2.4.1 Purpose

The separate modules are eventually integrated to form the whole VR system. The VR application is then evaluated based on several criteria: the objectives in all modules are met, the user's interactions, and the VR hardware input controls.

2.4.2 Activities and Tools

Four key activities in this stage: Functional Evaluations, UI/UX Evaluations, Version Control and Archives, and Installation/Deployment. Tools to be used at this stage include heuristic evaluation, testing cases, etc.

2.5 Stage 5: Project Closure

2.5.1 Purpose

This stage ensures the design requirements are met and the VR application continues to perform as per the specifications. Although the deployed version is now launched to the market, any issues reported by the end users must still be handled responsibly by the designer and developer.

2.5.2 Activities and Tools

There are two key activities at this stage, namely Documentation and System Upgrade.

3 RESULTS

3.1 User survey

We conducted a user survey of our proposed SPAT-VR framework with 4 questions:

- Q1: Has the user learned any existing VR project management framework other than SPAT-VR? (Options: Yes; No)
- Q2: SPAT-VR provides holistic information for VR project management? (Options: strongly disagree; disagree; neutral; agree; strongly agree)
- Q3: SPAT-VR is an easy-to-extend framework for VR project management? (Options: strongly disagree; disagree; neutral; agree; strongly agree)
- Q4: SPAT-VR is useful for my future VR project management? (Options: strongly disagree; disagree; neutral; agree; strongly agree)

In total 10 users completed the user survey. 6 users have >3 years of experience in VR project development and management while the rest 4 have ≤3 years of experience. The survey results showed that SPAT-VR is perceived as a novel, holistic, extensible and useful framework for VR project management (Figure 2).

4 CONCLUSION

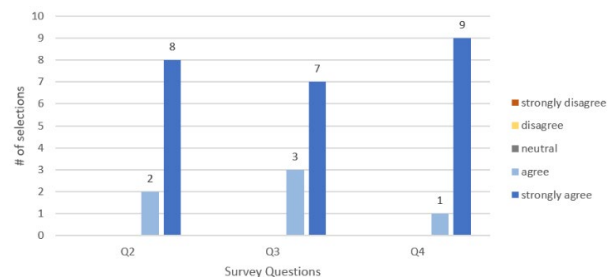


Figure 2: Survey result on SPAT-VR.

Our proposed a holistic and extensible SPAT-VR framework which can be extended to cover the whole lifecycle of VR projects. A user survey has been conducted. The SPAT-VR framework has also been applied to guide a few VR projects which have been successfully completed.

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REFERENCES

- [1] Mujber T S, Szecsi T, Hashmi M S J. Virtual reality applications in manufacturing process simulation. *Journal of Materials Processing Technology*, 2004, 155/156: 1834–1838.
- [2] Gallagher A G, Ritter E M, Champion H, Higgins G, Fried M P, Moses G. Virtual reality simulation for the operating room: proficiency-based training as a paradigm shift in surgical skills training. *Annals of Surgery*, 2005, 241: 364.
- [3] Thu Aung Ba R, Cai Y, Guan Y. Augmented reality simulation of cardiac circulation using APPLearn (heart). 2018 IEEE International Conference on Artificial Intelligence and Virtual Reality (AIVR), 2018, pp. 241-243.